Lecture 1 - Complexity and Algorithm Basics

Course: Algorithm Design and Analysis (ADA101)
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Summary

A compact lecture note covering algorithm definition, input size, time complexity, Big O notation, and a small comparison table.

Learning goals

- Define what an algorithm is.
- Explain input size and basic operation counting.
- Use Big O notation for simple loops.
- Compare constant, logarithmic, linear, and quadratic growth.

Core concepts

- An algorithm is a finite sequence of steps that transforms input into output.
- The same task can be solved by several algorithms with different time and memory costs.
- Big O describes the upper growth rate as input size becomes large, not the exact running time on one computer.

Example

- A loop that checks every item in a list of n elements performs n checks, so the running time grows linearly.
- A nested loop over all pairs of n items performs about n squared comparisons, so it grows quadratically.

Quick reference

Growth	Example	Practical meaning
O(1)	Read first item	Does not grow with input
O(log n)	Binary search	Adds one step when data doubles
O(n)	Linear scan	Doubles when input doubles
O(n^2)	All-pairs comparison	Becomes slow quickly